

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO1: Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems.(BL2,BL3)

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Case Study

Duration: 2hrs

1.A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. (5 Marks)

2.A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. (5 Marks)

3.An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. (5 Marks)

4.A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. (5 Marks)

5.A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. (5 Marks)

6.A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. (5 Marks)

7.A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance.(5 Marks)

8.A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. (5 Marks)

9.A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. (5 Marks)

10.A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. (5 Marks)

Code of Conduct:

I T.Chaitra(192425217) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:T.chaitra

RUBRICS FOR CASESTUDY

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

1. RL-Based Warehouse Robot

Definition

A Reinforcement Learning (RL)-based warehouse robot learns the optimal way to pick, pack, and transport items by interacting with the warehouse environment. It improves its performance through rewards and penalties while avoiding obstacles and minimizing delays.

MDP Components

- **States:** Robot location, package location, battery level, obstacle positions, delivery status.
- **Actions:** Move Forward, Turn Left, Turn Right, Pick Item, Pack Item, Deliver Item, Recharge.
- **Environment:** Warehouse containing shelves, packages, obstacles, and charging stations.
- **Policy:** Choose the safest and shortest path to complete deliveries efficiently.

Reward Function

- +20 → Successful delivery
- +10 → Correct item picked
- -10 → Collision with obstacle
- -5 → Wrong item picked
- -2 → Delay or unnecessary movement
- +5 → Efficient battery usage

Analysis

The reward function balances **speed, accuracy, and safety**. Positive rewards encourage successful deliveries, while penalties discourage collisions and delays. Over time, the robot learns the most efficient routes and reduces travel time.

Conclusion

The RL-based warehouse robot improves productivity, reduces delivery time, minimizes errors, and enhances warehouse automation.

2. Movie Recommendation System

Definition

A streaming platform uses Reinforcement Learning to recommend movies based on a user's viewing history, ratings, and preferences. The system continuously learns from user feedback to improve recommendations.

RL Components

- **States:** User profile, watch history, favorite genres, previous ratings.
- **Actions:** Recommend a movie, recommend a TV show, or recommend a new genre.
- **Environment:** Streaming platform and user interactions.
- **Policy:** Recommend content that maximizes user satisfaction.

Reward Function

- +10 → Movie watched completely
- +5 → Positive rating
- +3 → User adds to watchlist
- -5 → Movie skipped
- -10 → Negative rating

Analysis

The system balances **exploration** (recommending new genres) and **exploitation** (recommending favorite genres). Exploration helps discover new interests, while exploitation improves user satisfaction.

Conclusion

The RL recommendation system increases user engagement, watch time, and customer satisfaction through personalized suggestions.

3. Smart Home Automation

Definition

A Smart Home Automation System uses Reinforcement Learning to automatically control lighting, heating, and cooling based on occupancy and weather conditions while reducing electricity consumption.

RL Components

- **States:** Room temperature, occupancy status, weather conditions, electricity usage.
- **Actions:** Turn lights ON/OFF, adjust AC temperature, switch heating ON/OFF.
- **Environment:** Smart home with sensors and appliances.
- **Policy:** Maintain user comfort while minimizing energy usage.

Reward Function

- +10 → Comfortable room temperature
- +8 → Energy saved
- +5 → Lights switched off in an empty room
- -10 → User discomfort
- -5 → High electricity consumption

Analysis

Reward shaping encourages both **comfort** and **energy efficiency**. The agent learns to switch appliances on only when needed, reducing electricity bills without affecting user comfort.

Conclusion

The RL system improves energy efficiency, reduces electricity costs, and provides a comfortable living environment.

4. Fraud Detection System

Definition

A financial institution uses Reinforcement Learning to identify fraudulent transactions in real time by learning patterns from previous transactions and customer behavior.

RL Components

- **States:** Transaction amount, transaction location, user history, device information.
- **Actions:** Approve transaction, reject transaction, flag transaction for verification.
- **Environment:** Banking transaction network.
- **Policy:** Maximize fraud detection while minimizing false alarms.

Reward Function

- +20 → Correct fraud detection
- +10 → Correct approval of a genuine transaction
- -20 → Fraud not detected
- -15 → False alarm (blocking a genuine customer)

Analysis

The agent learns from transaction outcomes. One challenge is **imbalanced data**, as fraud cases are much rarer than normal transactions. Delayed confirmation of fraud also makes learning more difficult.

Conclusion

The RL system improves fraud detection accuracy, reduces financial losses, and enhances banking security.

5. Smart Grid Energy Management

Definition

A Smart Grid uses Reinforcement Learning to distribute electricity efficiently, reduce peak load demand, and ensure a reliable power supply.

MDP Components

- **States:** Electricity demand, available power, renewable energy generation, time of day.
- **Actions:** Increase power supply, decrease power supply, store energy, distribute renewable energy.
- **Environment:** Smart electrical grid with consumers and power sources.
- **Policy:** Balance electricity demand and supply efficiently.

Reward Function

- +15 → Balanced electricity distribution
- +10 → Reduced peak load
- +5 → Efficient use of renewable energy
- -15 → Power outage
- -10 → Energy wastage

Analysis

The RL agent learns to distribute electricity based on demand. It faces challenges such as changing weather, varying electricity consumption, and real-time decision-making. Reward shaping encourages reliable and efficient power distribution.

Conclusion

The RL-based Smart Grid improves energy efficiency, reduces power outages, minimizes wastage, and supports sustainable electricity management.